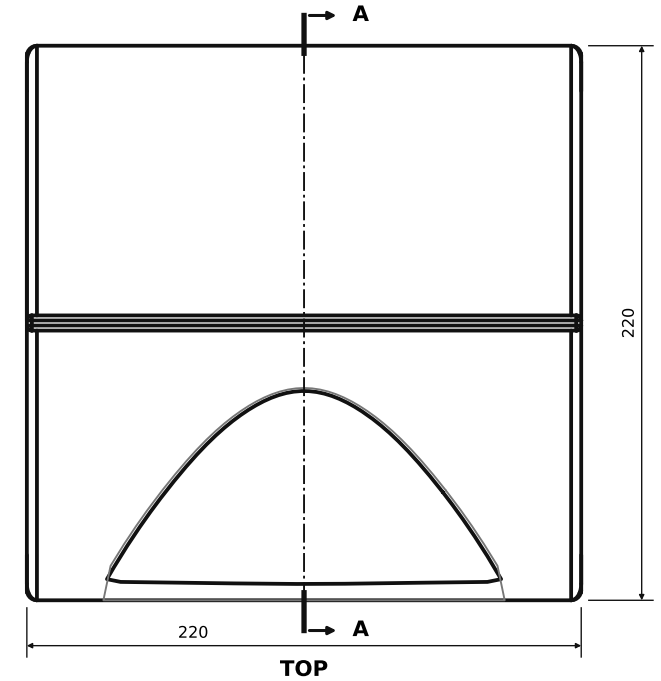
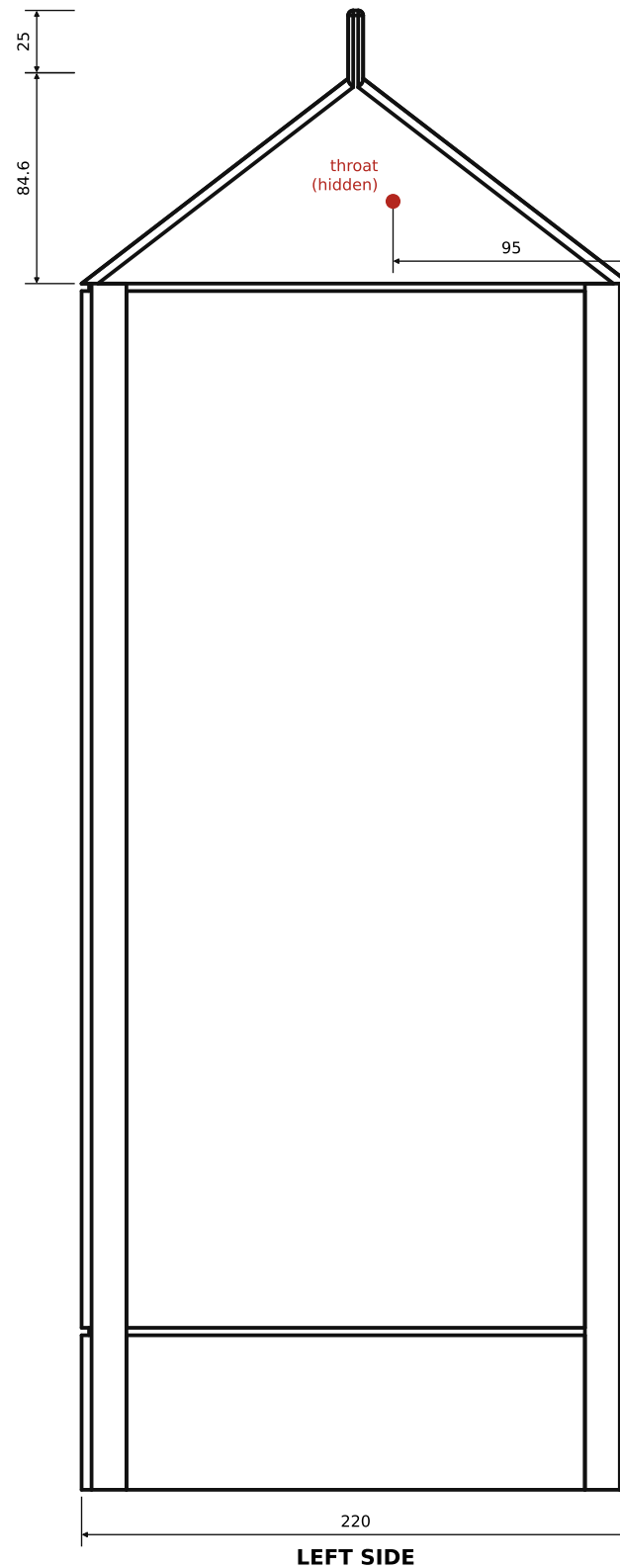
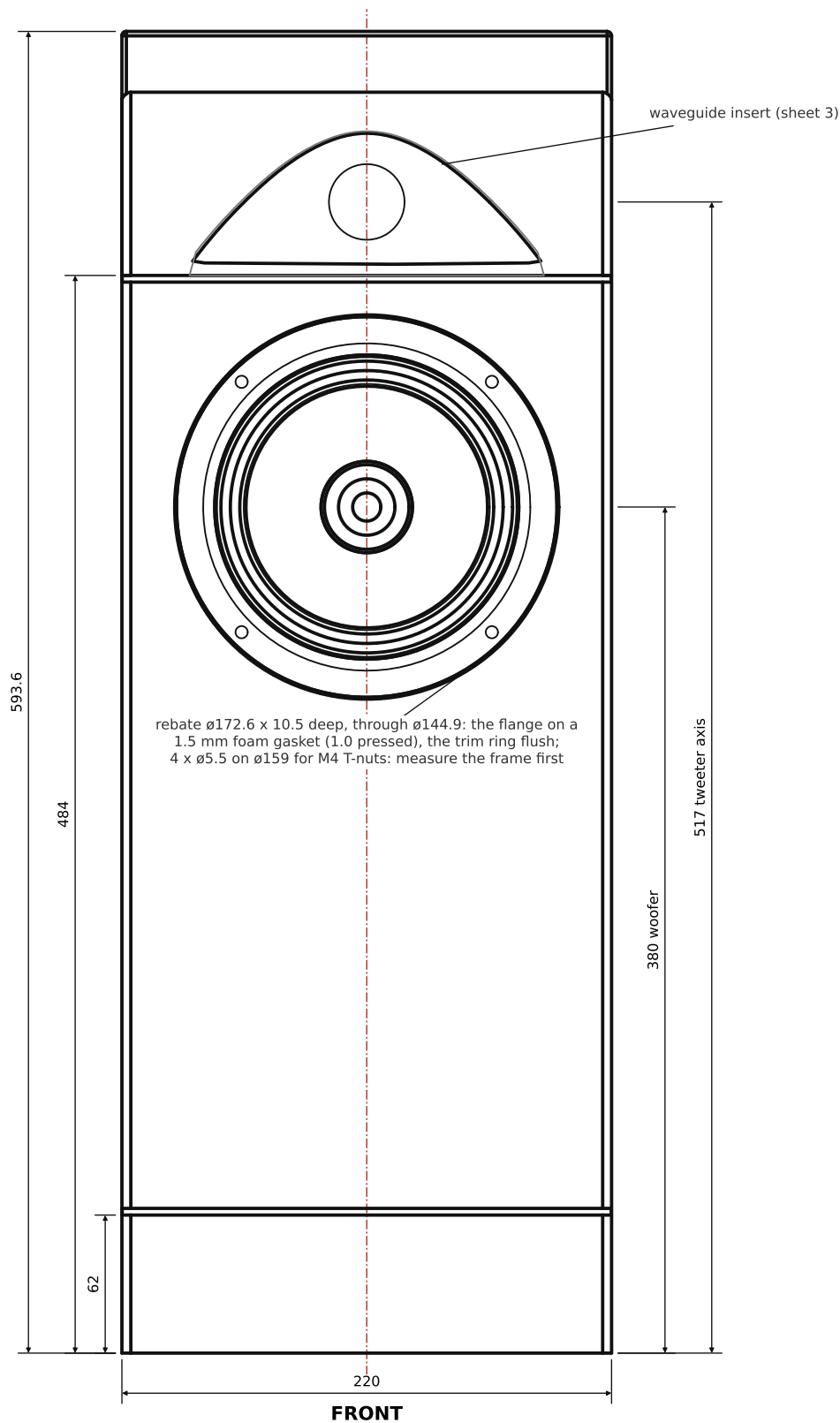




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

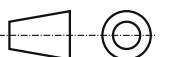
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 4 mm, the fin's edges 2 mm. 3 x 3 mm shadow lines at z 62 and z 481.
2. The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 10.5 for a 6.5 flange; measure each flange before cutting. M4 x 20 button heads 1.6 high or less (ISO 7380 is 2.2) into M4 T-nuts pressed into the inside face before the glue-up, barrels 7 or less. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer $\varnothing 171$ / $\varnothing 147 \times 3$, a channel 10 wide x 2 deep in its back over the screws' circle) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner $\varnothing$ is the surround at its glue line + 2: measure the drivers and print the rings last.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 517, 95 behind the front face. The shape is set by simulation (fab/out-bookshelf/acoustics/waveguide).
4. The back (the amplifier) is drawn on sheet 4 with the wiring; the Nutrition Facts are printed on the right side.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk bookshelf | General arrangement

Front, left side and top at 1:3. Heights from the floor; depths from the front face.

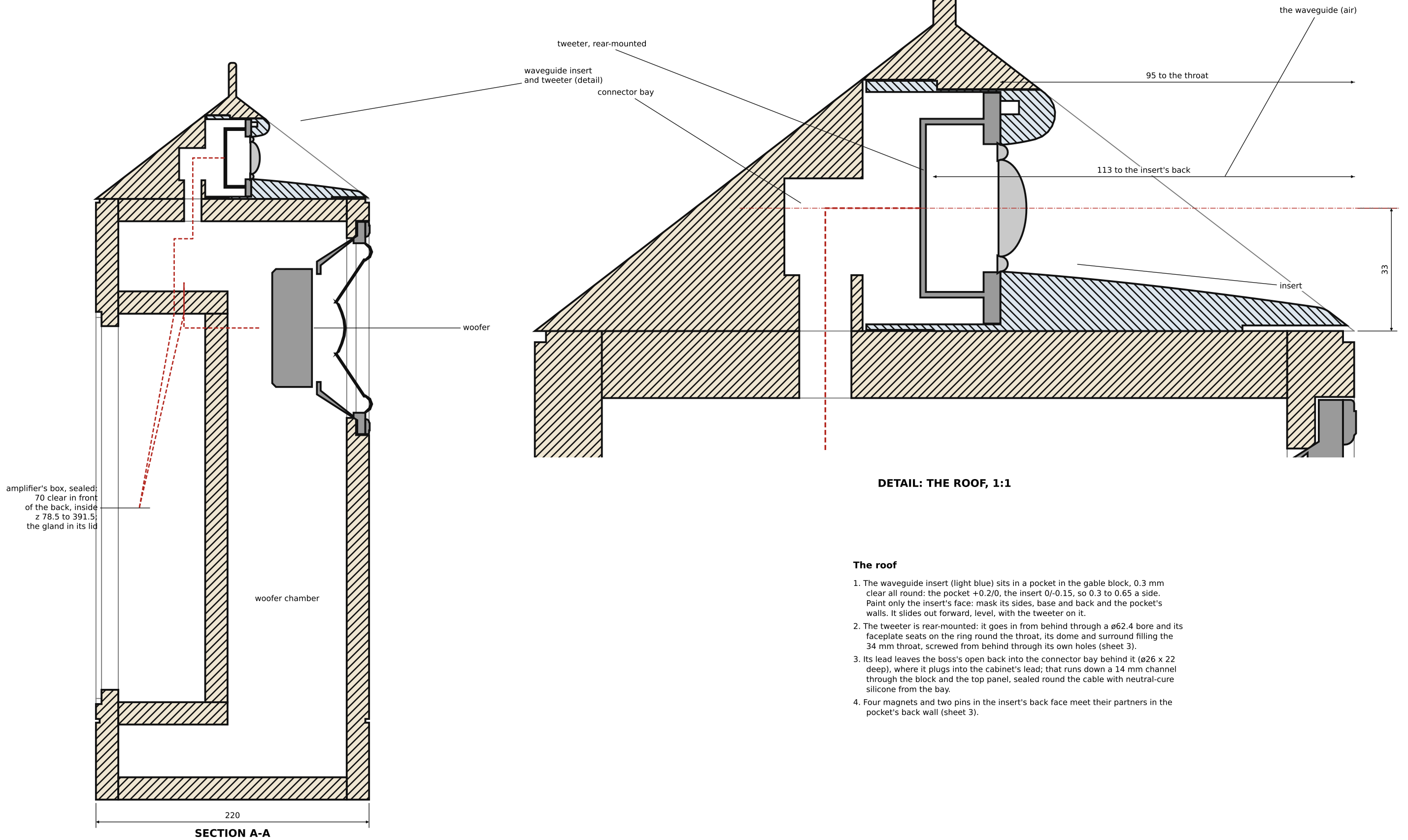
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



1 of 6

EM-BS-001 rev C



red dashed: the cables' runs to the amplifier (sheet 4)

DETAIL: THE ROOF, 1:1

The roof

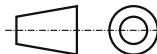
1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket $+0.2/0$, the insert $0/-0.15$, so 0.3 to 0.65 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: it goes in from behind through a $\varnothing 62.4$ bore and its faceplate seats on the ring round the throat, its dome and surround filling the 34 mm throat, screwed from behind through its own holes (sheet 3).
3. Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 26 \times 22$ deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

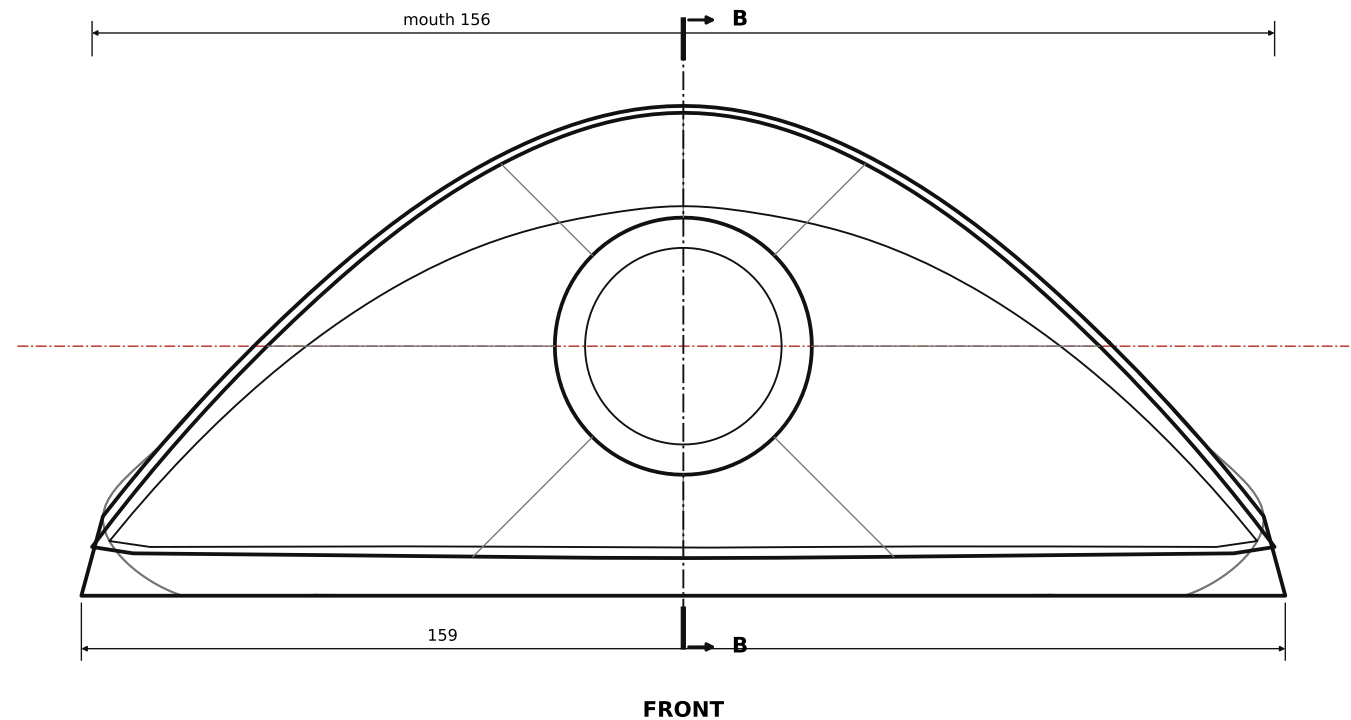
earmilk bookshelf | Section on the centreline

Section A-A on the centre plane (x 110), from the left, at 1:3; the roof at 1:1. The insert hatched the other way.

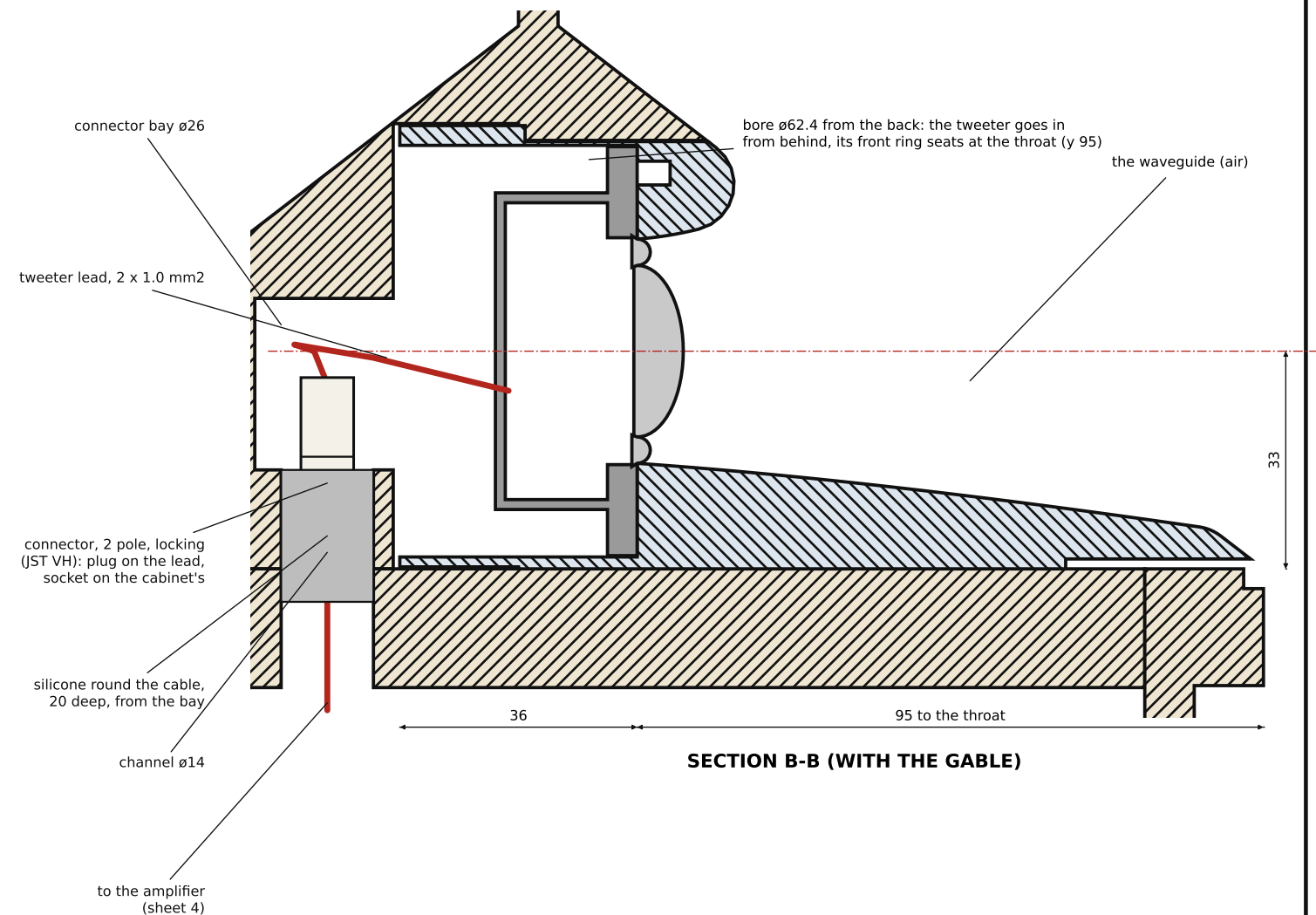
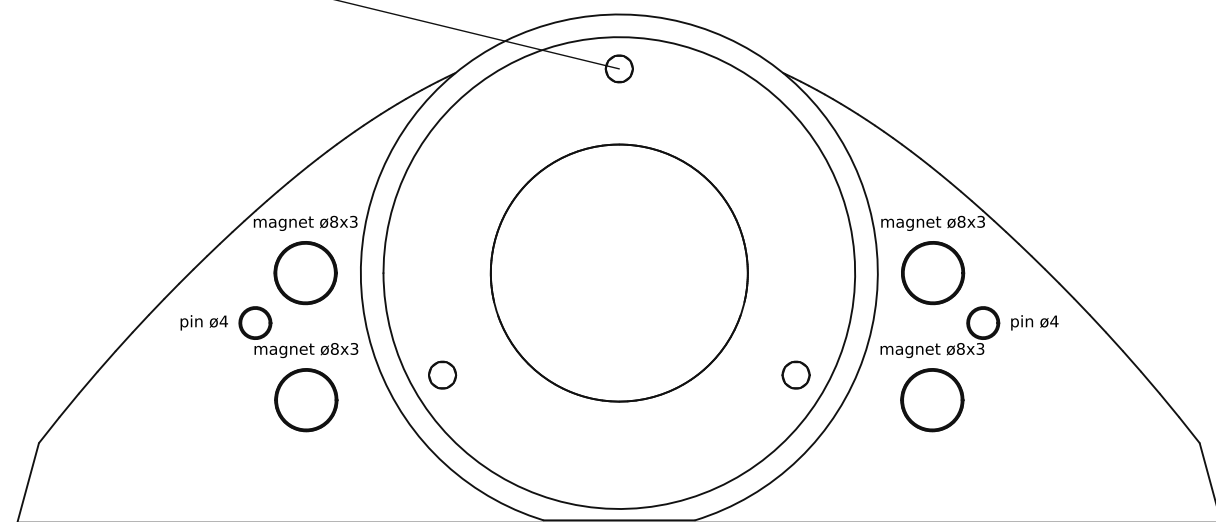
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.





3 x $\varnothing 3.6 \times 5$ holes on $\varnothing 54$ in the seat, one at the top, 120° apart, for M2.5 x 4 knurled brass inserts bonded with epoxy;
M2.5 x 8 button heads (ISO 7380, A2) through the faceplate: measure it first



Fitting the tweeter and the insert

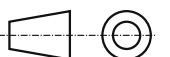
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Bond 3 M2.5 x 4 knurled brass inserts into the seat with epoxy ($\varnothing 3.6 \times 5$ holes on $\varnothing 54$; a heat-set insert will not melt into cured resin). Pass the tweeter in from behind through the $\varnothing 62.4$ bore, dome first, seat its faceplate on a 0.5 mm foam gasket and screw it to the inserts with M2.5 x 8 button heads (ISO 7380, A2) through its own holes. Measure first: the heads need 0.5 clear of the body (the circle at least the body + 5.7).
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
4. The cabinet's lead ends in the socket 185 mm above the top panel's underside. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay. Then seal the channel round the cable with 20 mm of neutral-cure silicone from the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0, the insert 0/-0.15 (0.3 to 0.65 a side); paint only the insert's face.
6. Service: pull it out by a ribbon loop glued in the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it.
7. The pocket (gable-block.step): the insert's outline + 0.3 a side (159.8 wide at its base), from the slope to its back wall at y 113.3; the boss's bore $\varnothing 69$ to y 132; the connector bay $\varnothing 26$ from y 131 to 153 on an axis at z 512; the channel $\varnothing 14$ at x 110, y 142 down to the top panel; internal corners R3 or less (the insert's match); dowel holes $\varnothing 10 \times 20$ in the base at (34, 125), (186, 125), (34, 186), (186, 186).
8. Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ± 41.4 across, 16.2 up; ± 41.5 across, 33.0 up; pins ± 48.1 across, 26.4 up. Magnets $\varnothing 8 \times 3$ in $\varnothing 8.2 \times 3.3$ holes, epoxied, poles marked; pins $\varnothing 4 \times 8$ bonded with epoxy in $\varnothing 4.1 \times 4.5$ in the insert, a slip fit in $\varnothing 4.2 \times 4.5$ in the wall (these holes are outside the title block's +0.2/-0).

earmilk bookshelf | Waveguide insert and tweeter mount

The insert at 1:1: front, section on the axis, back; the pocket; fitting and service.

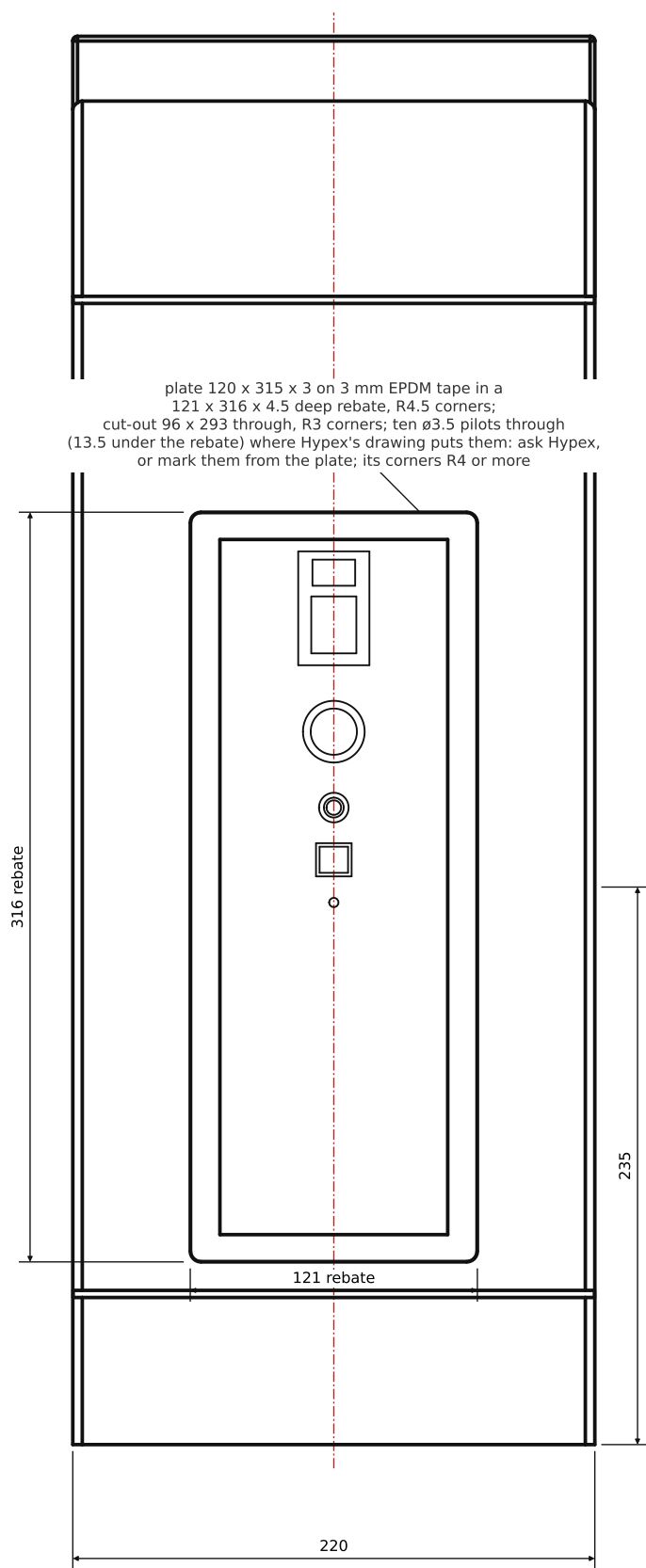
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes +0.2/-0, prints ± 0.15 . Check bought parts first.



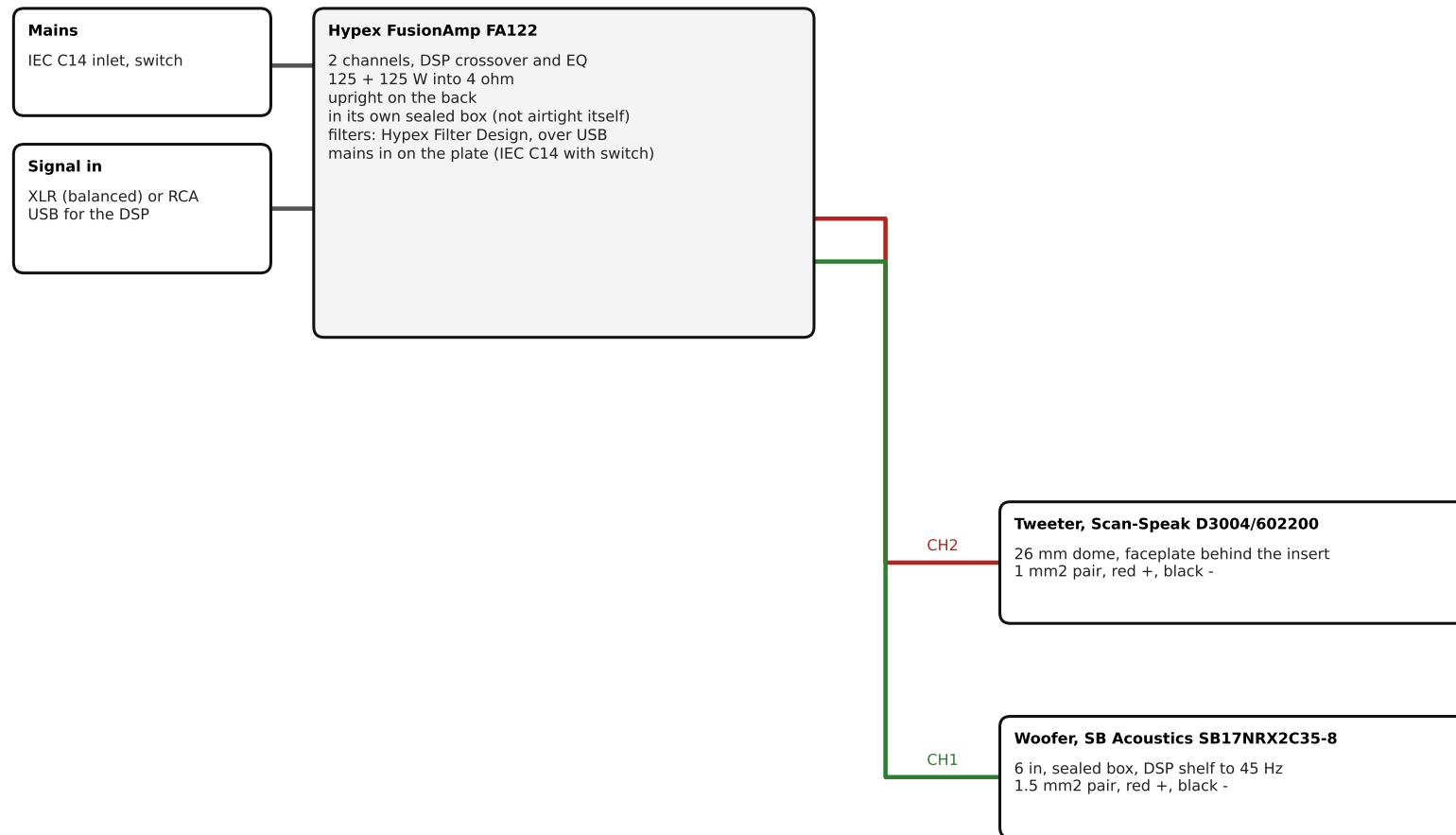
3 of 6

EM-BS-003 rev C



BACK

The Nutrition Facts go on the right side: 150 x 147, top at z 420, centred across its depth, 35 from each vertical corner.



Runs, connectors and seals

1. Woofer (CH1): from the amplifier box's gland to the woofer's terminals, 0.4 m of 1.5 mm2.
2. Tweeter (CH2): from the gland up behind the woofer's magnet to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert.
3. Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3): the box is sealed, and the insert's pocket is open to the room through its seam.
4. Round-sheathed cable throughout (H05VV-F 2 x 1.5 and 2 x 1.0 mm2, about 8.0 and 6.4 across), so the gland's seal grips it; splice any extension of Hypex's harness inside the amplifier's box (crimped butt splices under heat-shrink).
5. The amplifier box's lid: an M20 x 1.5 long-thread (15 mm) gland with a two-hole seal for the two pairs, in a ø20.5 hole 144 from the left side's inner face and 43 from the lid's front edge, its lock nut in a ø32 x 8 counterbore under the lid; seal it.
6. The crossover (2.4 kHz, LR4), the woofer's shelf to 45 Hz (+8.3 dB) and the delays are set in the DSP. Wire red to + throughout.

earmilk bookshelf | Back and wiring

The back at 1:3 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

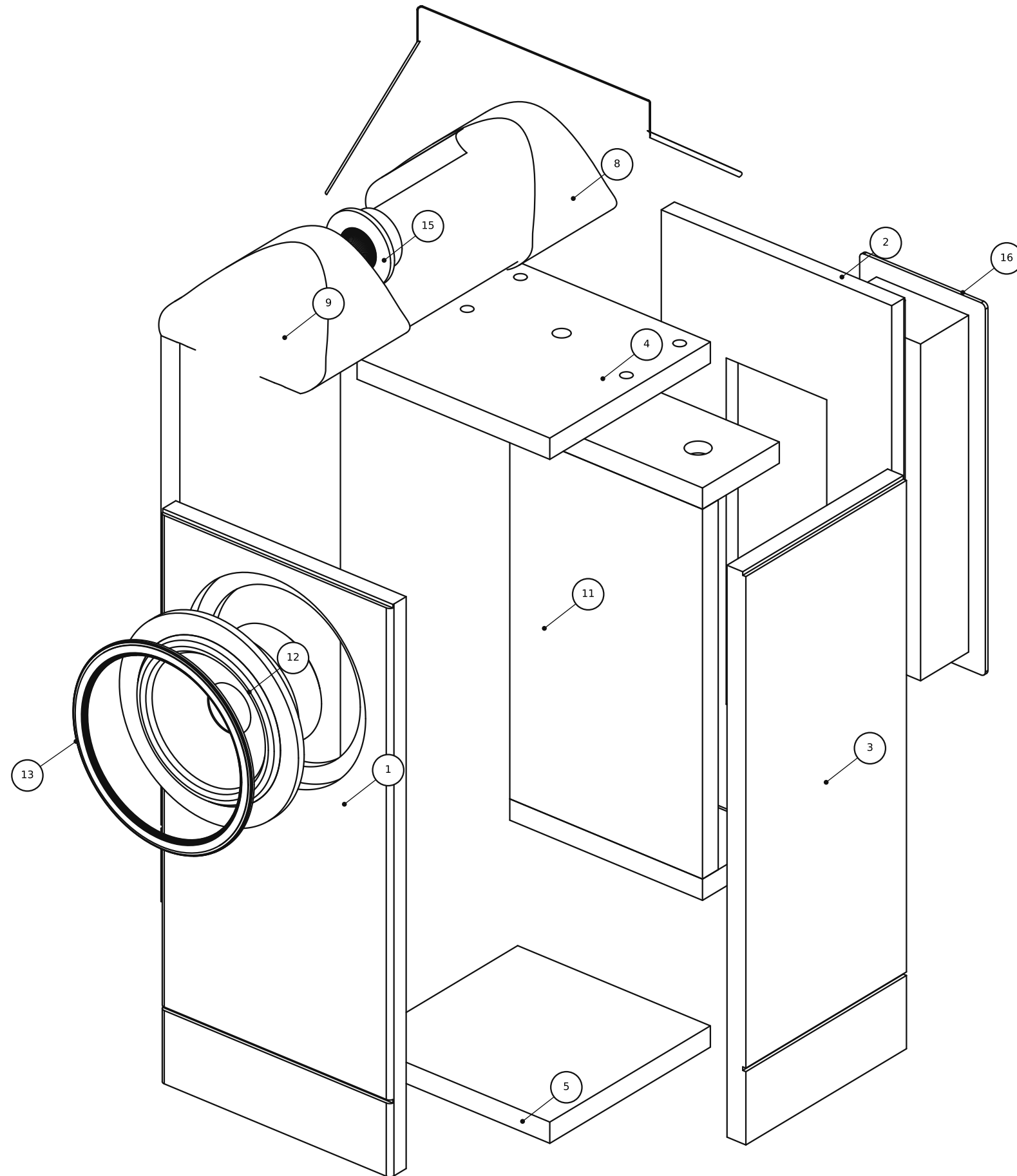
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.



4 of 6

EM-BS-004 rev C



Parts

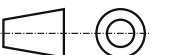
- ① **Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cutout and rebate
- ② **Back x1**
18 mm birch: the amplifier's cutout and rebate
- ③ **Sides x2**
18 mm birch; the Facts printed on the right one
- ④ **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑧ **Gable block x1**
laminated birch or printed whole: the insert's pocket and the cable channel
- ⑨ **Waveguide insert x1**
printed (SLA, MJF) or CNC from solid; magnets, pins
- ⑪ **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- ⑫ **Woofer x1**
SB Acoustics SB17NRX2C35-8, 6 in coated paper
- ⑬ **Trim ring x1**
printed, satin black or gunmetal, over the frame and its screws
- ⑮ **Tweeter x1**
Scan-Speak Illuminator D3004/602200, faceplate behind the insert
- ⑯ **Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

earmilk bookshelf | Exploded view and parts

Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

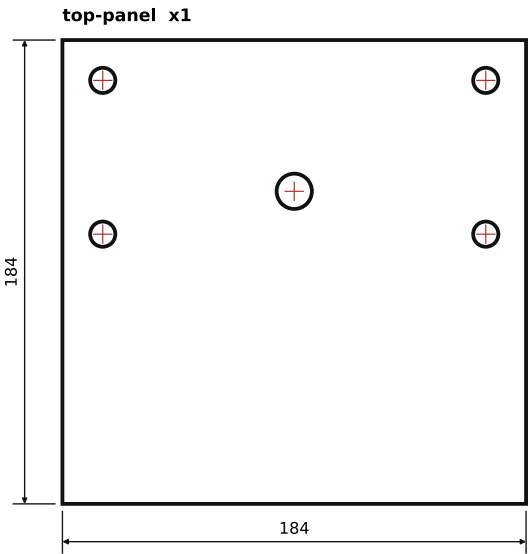
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.

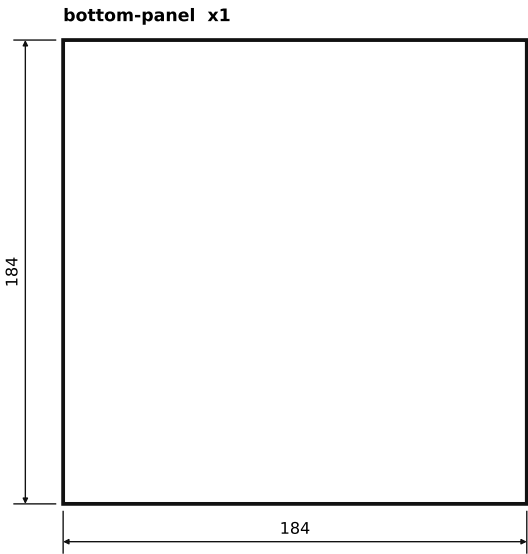


5 of 6

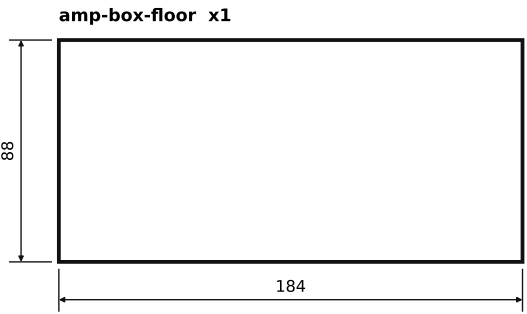
EM-BS-005 rev C



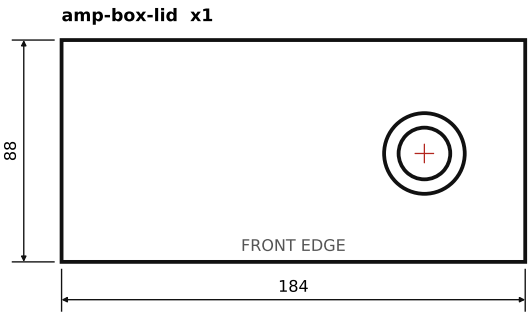
ø14 at (92, 124): cut inside
ø10 at (16, 107): drill ø10, 10 deep
ø10 at (168, 107): drill ø10, 10 deep
ø10 at (16, 168): drill ø10, 10 deep
ø10 at (168, 168): drill ø10, 10 deep
upper face up (front edge at the bottom of the drawing); dowels
register the gable block; the tweeter cable's hole is sealed with
silicone from the bay after wiring



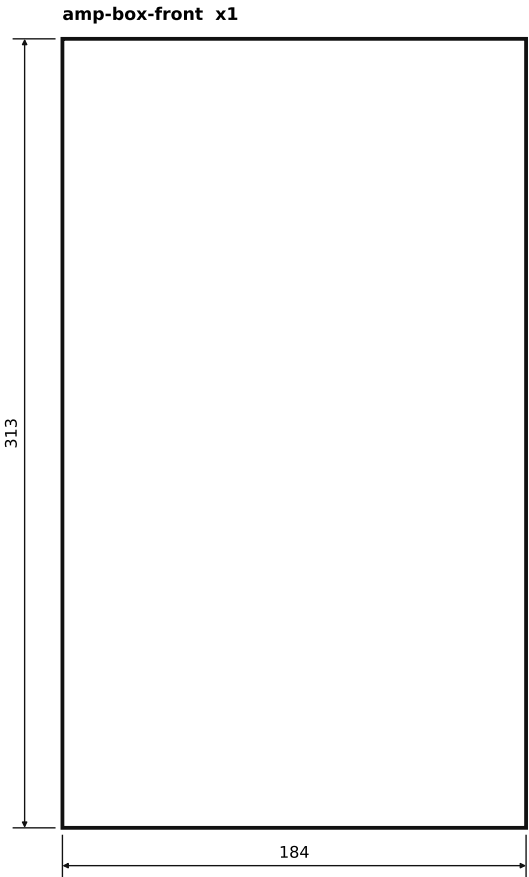
either face



the amplifier box's floor, z 60.5 to 78.5, against the back



ø20.5 at (144, 43): cut inside
ø32 at (144, 43): pocket 8 mm deep from the underside, concentric
with the gland's hole (cut the hole, turn the panel over, centre
on it; before the lid goes in)
the amplifier box's lid, z 391.5 to 409.5, its front edge (the
bottom of the drawing) over the box's front; the gland's ø20.5
hole 144 from the left, 43 from the front edge, its nut in a ø32 x
8 counterbore from the underside; sealed after wiring



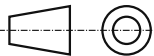
the amplifier box's front, between floor and lid, 70 in front of
the back's inner face; glue and seal all round

earmilk bookshelf | Panel details

At 1:3: each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm birch.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



6 of 6

EM-BS-006 rev C